ÖZYEĞİN ÜNİVERSİTESİ

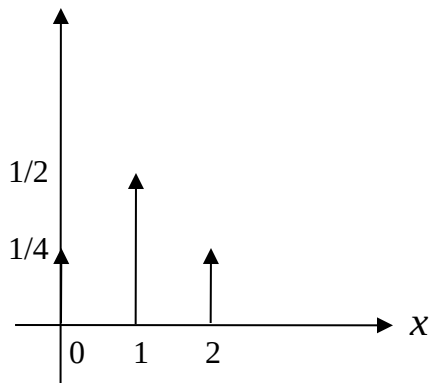
MATH 215
Probability and Random Variables

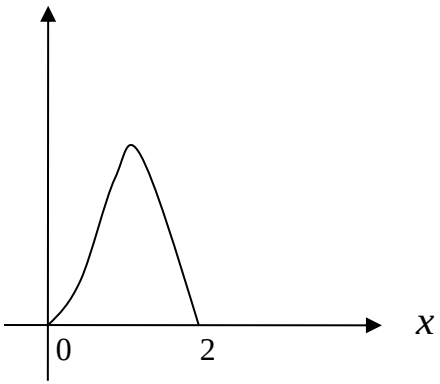
ENİS KAYIŞ

Parts of these notes are based on Murat Uysal's ECE316 lecture notes.

Random Variables

- **Definition:** A random variable is a real-valued function of the elements of a sample space, S. Given an experiment E, the random variable X maps each possible outcome $s \in S$ to a real number $X(s)$ as specified by some rule.
- If X takes on a finite or countably infinite number of values, X is called as a discrete random variable.
- If X takes on uncountably infinite number of values, X is called as a continuous random variable.





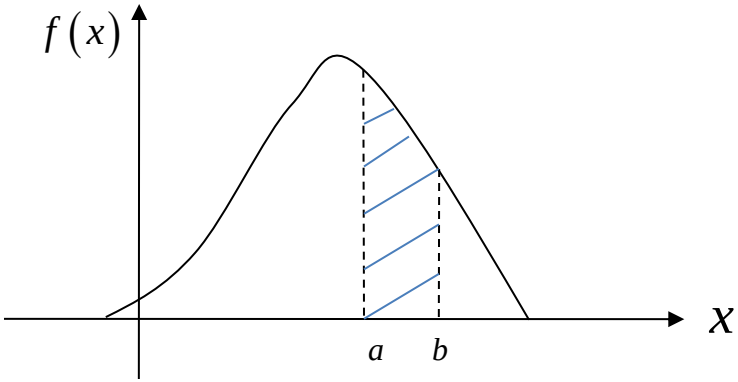
Continuous Random Variables

- **Definition:** The probability density function (pdf) $f(.)$ of a continuous random variable X evaluated at the point x is

$$f(x) = \lim_{\varepsilon \to 0} \frac{P(x \le X \le x + \in)}{\in}$$

This corresponds to the probability that X lies in an infinitesimal interval about the point $X=x$, normalized by the length of the interval.

- Let $A=[a, b]$



$$P(X \in A) = \int_A f(x)dx$$

$$P(a \le X \le b) = \int_a^b f(x)dx$$

Continuous Random Variables (cont'd)

- Recall that the cumulative distribution function (cdf) specifies the random variable is less than or equal to *x*.

$$F(x) = P(X \le x)$$

- Properties of cdf

$$F(-\infty) = 0 \quad F(\infty) = 1$$

$$0 \le F(x) \le 1$$

For $a<b$, $\quad F(b) \le F(a)$

For $a<b$, $\; P(a \le X \le b) = F(b) - F(a)$

Continuous Random Variables (cont'd)

- Relation between pdf and cdf

$$P\left(x \le X \le x+\varepsilon\right) = F\left(x+\varepsilon\right) - F\left(x\right)$$

$$f\left(x\right) = \lim_{\varepsilon \to 0} \frac{P\left(x \le X \le x+\varepsilon\right)}{\varepsilon}$$

$$= \lim_{\varepsilon \to 0} \frac{F\left(x+\varepsilon\right) - F\left(x\right)}{\varepsilon} = \frac{dF\left(x\right)}{dx}$$

- Alternatively, we have

$$F\left(x\right) = P\left(X \le x\right) = \int_{-\infty}^{x} f\left(t\right)dt$$

- It can be readily checked that

$$P\left(-\infty < X < \infty\right) = F\left(\infty\right) - F\left(-\infty\right) = \int_{-\infty}^{\infty} f\left(x\right)dx = 1$$

Expectation and Variance

- Expected value of continuous random variable X

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

- If X is a random variable with mean $\mu_X = E(X)$, the variance of X is

$$\mathrm{Var}(X) = E\left[(X - \mu_X)^2\right] = E\left(X^2\right) - \mu_X^2$$

Expectation and Variance (cont'd)

- **Example:** Find the expected value and variance of X whose pdf is given by

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \int_0^1 2x^2 dx = 2/3$$

$$E(X^2) = \int_0^1 2x^3 dx = 1/2$$

$$Var(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

Expectation and Variance (cont'd)

- **Proposition:** Assume X is a continuous random variable. The expected value of $g(X)$ where $g(.)$ is a real-valued function is given by

$$E\left(g\left(X\right)\right)=\int_{-\infty}^{\infty} g\left(x\right)f\left(x\right)dx$$

- Expectation is a linear operation.

$$E\left(aX+b\right)=aE\left(X\right)+b$$

- Variance is not a linear operation.

$$Var\left(aX+b\right)=a^2Var\left(X\right)$$

Continuous Random Variables (cont'd)

- **Theorem:** X is a continuous random variable with pdf $f_X(x)$. Suppose that $g(x)$ is a strictly monotonic (increasing or decreasing), differentiable and thus continuous function of x. Let Y denote another random variable as a function of X as $Y=g(X)$. The pdf of Y is given by

$$f_Y(y) = f_X\left[g^{-1}(y)\right]\left|\frac{dg^{-1}(y)}{dy}\right|$$

- **Proof:** Assuming $g(x)$ is an increasing function

$$F_Y(y) = P\{Y \le y\} = P\{g(X) \le y\} = P\left\{X \le g^{-1}(y)\right\} = F_X\left(g^{-1}(y)\right)$$

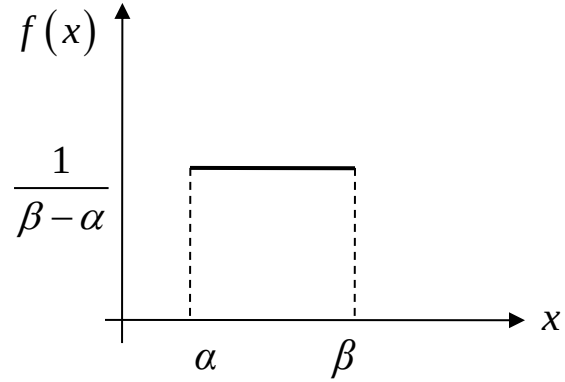
Differentiation gives

$$f_Y(y) = \frac{dF_Y(y)}{dy} = f_X\left(g^{-1}(y)\right)\frac{dg^{-1}(y)}{dy}$$

Uniform Random Variable

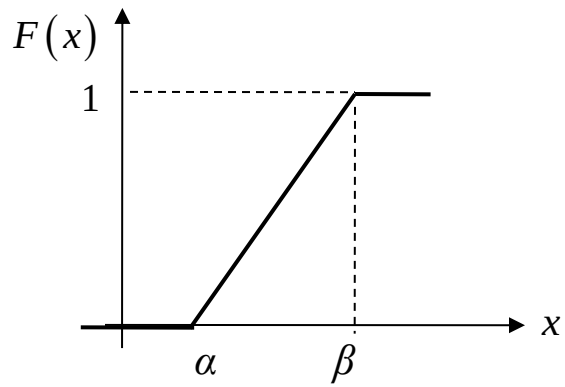
- X is a uniform random variable on the interval (α,β) if its pdf is given by

$$f(x)=\begin{cases}\dfrac{1}{\beta-\alpha}, & \alpha<x<\beta\\ 0, & \text{otherwise}\end{cases}$$



- The cdf of X is given by

$$F(x)=\begin{cases}0, & x\le\alpha\\ \dfrac{x-\alpha}{\beta-\alpha}, & \alpha<x<\beta\\ 1, & x\ge\beta\end{cases}$$



Uniform Random Variable (cont'd)

- Expected value

$$E\left(X\right)=\int_{\alpha}^{\beta}\frac{x}{\beta-\alpha}dx=\frac{1}{\beta-\alpha}\frac{x^2}{2}\bigg|_{\alpha}^{\beta}=\frac{\beta^2-\alpha^2}{2\left(\beta-\alpha\right)}=\frac{\beta+\alpha}{2}$$

- Variance

$$E\left(X^2\right)=\int_{\alpha}^{\beta}\frac{x^2}{\beta-\alpha}dx=\frac{1}{\beta-\alpha}\frac{x^3}{3}\bigg|_{\alpha}^{\beta}=\frac{\beta^3-\alpha^3}{3\left(\beta-\alpha\right)}=\frac{\beta^2+\alpha\beta+\alpha^2}{3}$$

$$Var\left(X\right)=\frac{\beta^2+\alpha\beta+\alpha^2}{3}-\frac{\left(\alpha+\beta\right)^2}{4}$$

$$=\frac{4\beta^2+4\alpha\beta+4\alpha^2-\left(3\alpha^2+6\alpha\beta+3\beta^2\right)}{12}=\frac{\left(\beta-\alpha\right)^2}{12}$$